SAYAK DUTTA

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PROFILE

Senior AI/ML Engineer with 10+ years of experience, I build AI that ships and stays up. Specialised in multi-agent orchestration (LangGraph, LangChain), hybrid RAG architectures, and real-time distributed systems on AWS and GCP. Rooted in data engineering, entity resolution, NLP pipelines, high-volume data systems, which gives me the infrastructure rigour most AI engineers skip. I work at the intersection of AI architecture and product delivery, partnering with product managers and business stakeholders to turn capability into outcomes: 60% latency reduction, 45% LLM cost savings, 99.9% production reliability across live systems.

PROFESSIONAL EXPERIENCE

Siaris GmbH, *AI Engineer* 2025 – Present | Germany

- Architected and shipped the company's core multi-agent LLM platform (LangGraph + LangChain) from scratch, now the primary production system powering workflows for 100+ daily active users, live within 3 months of joining.
- Redesigned agent orchestration on AWS Lambda + ECS; cut median response latency from 12s to under 6s, directly driving a 38% increase in weekly active usage.
- Designed hybrid RAG with HNSW vector retrieval + semantic re-ranking; LangSmith-based evaluation harness measured a 42% improvement in context relevance post-rollout.
- Architected the production safety layer for enterprise LLM applications, designing permission-aware retrieval, hallucination evaluation workflows, PII-isolated document access, and policy enforcement to balance answer quality, security, and compliance while maintaining 99%+ production reliability.
- Cut LLM inference costs by 45% through chunking optimisation, Redis retrieval caching, and intelligent model routing, zero quality degradation.
- Architected an MCP-based document integration layer that exposed enterprise PDF and Excel data through standardized MCP tools for a LangGraph multi-agent platform, enabling agents to search, retrieve, and extract proprietary knowledge without custom source-specific integrations.
- Owned AI architecture from product discovery through production deployment, working directly with founders and enterprise customers to translate business requirements into scalable multi-agent workflows, retrieval strategies, and deployment decisions.

Torto Labs Pvt Ltd., *Senior Machine Learning Engineer* 2021 – 2025 | India (Remote)

- Architected and shipped an enterprise RAG platform combining semantic vector retrieval, structured SQL access, and LLM reasoning to power natural-language analytics over 1M+ records/month on AWS.
- Slashed analyst lookup latency from 60s to under 20s (ElastiCache + async inference); grew product adoption from 22% to 61% within 6 months.
- Established the organization's production LLM evaluation framework using LangSmith, defining retrieval recall, grounding accuracy, latency SLAs, and release gates that became the standard validation process for every AI feature.
- Architected event-driven inference pipelines on AWS MSK (Kafka) + ElastiCache (Redis), 99.9% uptime under variable production load.
- Established CI/CD via GitHub Actions + AWS CodeDeploy; cut deployment cycle time by 60%+ and moved the team to weekly release cadence.
- Led technical planning across engineering, product, and customer stakeholders, driving architecture decisions, mentoring engineers on production RAG systems, and defining engineering standards for evaluation and deployment.

Berylls GmbH, *Machine Learning Engineer* 2023 – 2023 | Germany

- Productionised collaborative filtering models into FastAPI endpoints on GCP Cloud Run, 100K+ prediction requests/month at p95 latency under 80ms.
- Optimized recommendation features through feature engineering, model tuning, and offline experimentation, increasing engagement by 25% and conversion by 15%, validated through A/B testing.
- Designed automated model monitoring and retraining pipelines with production drift detection, eliminating manual health checks while improving operational reliability and deployment confidence.
- Containerised the full ML stack with Docker + GCP Artifact Registry; reduced MLOps overhead by 80% with fully reproducible deployments.

Nailbiter Inc., *Senior Data Analyst* 2020 – 2021 | India

- Architected scalable NLP pipelines for large-scale consumer intelligence, transforming unstructured feedback into production-ready sentiment, entity, and thematic analytics that informed business decision-making.
- Automated text classification and entity tagging, replacing manual workflows; 20% targeting accuracy improvement and 30% faster data processing.
- Built distributed data pipelines aggregating multi-source consumer intelligence into unified analytics datasets, reducing campaign planning time from days to hours.

eClerx Services Ltd., *Analyst* 2018 – 2020 | India

- Designed entity resolution and fuzzy matching systems (Levenshtein + phonetic) processing 500K+ records/month, 35% accuracy improvement across client product catalogues.
- Re-architected ETL validation and reporting pipelines, improving data quality, reducing operational risk, and shrinking reporting latency from three days to same-day delivery.

EDUCATION

Bachelor of Engineering– Mechanical Engineering, *University of Mumbai* 2012 – 2016 | Mumbai, India
Grade: First Division

SKILLS

AI / LLM: LLMs (OpenAI, Anthropic, OSS), RAG & Hybrid Retrieval, Multi-Agent Systems, LangChain, LangGraph, LangSmith, FAISS, pgvector, MCP, Embeddings, Prompt Engineering, Model Evaluation

ML / Data: PyTorch, Hugging Face Transformers, Scikit-learn, NLP, Recommendation Systems, Time-Series, Feature Engineering

Full-Stack: Python, TypeScript / JavaScript, FastAPI, Node.js, Next.js, REST APIs, WebSockets, PostgreSQL

Cloud: AWS (Lambda, ECS, S3, RDS, MSK, ElastiCache, CodeDeploy), GCP (Cloud Run, Artifact Registry, Monitoring)

Infrastructure: Docker, Kubernetes, Kafka, Redis, Celery, GitHub Actions CI/CD, Serverless, MLOps, Model Monitoring

PROJECTS

ResearchRAG - Academic PDF Research Assistant
- Architected a RAG pipeline with custom chunking strategy (sliding window + section-aware splitting) handling complex academic PDF structure, figures, citations, methodology sections, more accurately than naive approaches.
- Built hybrid retrieval using FAISS dense search, BM25 lexical ranking, and LangChain orchestration with LangSmith evaluation, generating grounded structured answers from 30-page research papers in under 30 seconds.
- Tech stack: FastAPI · Next.js · TypeScript · FAISS · BM25 · LangChain · LangSmith · PyMuPDF · OpenRouter OSS

AskPostgres - Natural Language to SQL for Business Intelligence
- Built a production NL-to-SQL pipeline with schema-aware validation, syntax verification, and SQL injection safeguards, preventing invalid queries from reaching execution while maintaining accurate LLM-generated SQL.
- Schema-level RBAC enforced at query generation time; cut analyst SQL-writing time by up to 80%.
- Tech stack: FastAPI · Next.js · TypeScript · PostgreSQL · pgvector · LangChain · OpenRouter

OutreachPilot - Async Campaign Automation & Analytics Platform
- Event-driven orchestration with Celery + Redis queues, decoupled send scheduling from delivery tracking; dead-letter queues + retry policies for fault tolerance.
- Custom pixel + redirect tracking middleware for open rates, CTR, and A/B variant performance at individual recipient level.
- Tech stack: FastAPI · Next.js · TypeScript · PostgreSQL · Celery · Redis · Docker · Resend API

TalkToTube - Long-Form Video Knowledge Retrieval
- Designed semantic transcript segmentation and retrieval pipelines for hour-long videos, enabling grounded question answering with timestamp-linked citations instead of fixed-window chunking.
- LangChain QA chain over FAISS vector store; answers cite source timestamps so users jump directly to the relevant segment.
- Tech stack: FastAPI · Next.js · TypeScript · PostgreSQL · FAISS · LangChain · Docker · YouTube Transcript API

LANGUAGES AND VISA

Languages:
English (Fluent), Hindi (Native), German (A2)

Visa / Work Authorization:
Indian Citizen; authorized to work without restrictions